

Problem

Design a series RLC type band-pass filter with cutoff frequencies of 10 kHz and 11 kHz. Assuming $C = 80$ pF, find R , L , and Q .

Step-by-step solution

Step 1 of 8

We have to design a bandpass filter to pass all the frequencies with in the band of frequencies

$$\omega_1 < \omega < \omega_2$$

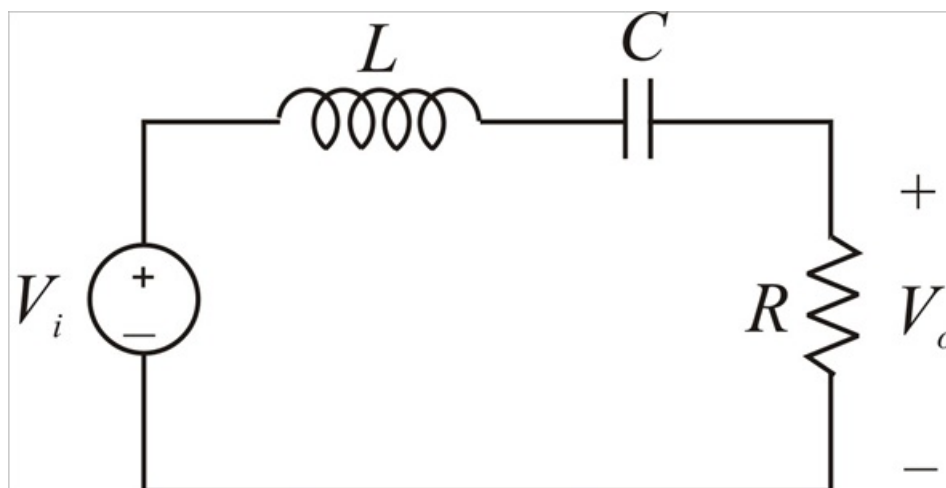
Here ω_1 = lower cutoff frequency

ω = center frequency

ω_2 = upper cutoff frequency

Step 2 of 8

Consider below series RLC circuit



Step 3 of 8

Transfer function of the above circuit

$$H(\omega) = \frac{V_o(\omega)}{V_i(\omega)}$$

$$H(\omega) = \frac{R}{R + j\omega L - \frac{j}{\omega C}}$$

$$H(\omega) = \frac{R}{R + j\left(\omega L - \frac{1}{\omega C}\right)}$$

Step 4 of 8

From the above equation $H(0) = 0$ and $H(\infty) = 0$

Hence this circuit acts like a bandpass filter

Center frequency of this filter is

$$\omega_o = \frac{1}{\sqrt{LC}}$$

Bandwidth of this filter $B = \omega_2 - \omega_1 = 2\pi \times 11 \times 10^3 - 2\pi \times 10 \times 10^3$

$$B = 2\pi \times 1 \times 10^3 \text{ rad/sec}$$

Step 5 of 8

Relation between center frequency and side band frequencies

$$\omega_o = \frac{\omega_1 + \omega_2}{2}$$

$$f_o = \frac{f_1 + f_2}{2}$$

$$f_o = \frac{10 \times 10^3 + 11 \times 10^3}{2}$$

$$f_o = 10.5 \text{ kHz}$$

$$\omega_o = 2\pi f_o$$

$$\omega_o = 2\pi (10.5 \times 10^3)$$

$$\omega_o = 65.97 \text{ krad/sec}$$

Step 6 of 8

Quality factor of the filter

$$Q = \frac{\omega_o}{\omega_2 - \omega_1}$$

$$Q = \frac{2\pi \times 10.5 \times 10^3}{2\pi \times 1 \times 10^3}$$

$$Q = 10.5$$

Step 7 of 8

$$\omega_o = \frac{1}{\sqrt{LC}}$$

$$L = \frac{1}{\omega_o^2 C}$$

Given $C = 80 \text{ pF}$

$$L = \frac{1}{(65.97 \times 10^3)^2 (80 \times 10^{-12})}$$

$$L = 2.872 \text{ H}$$

Step 8 of 8

Quality factor

$$Q = \frac{\omega_o L}{R}$$

$$R = \frac{\omega_o L}{Q}$$

$$R = \frac{(65.97 \times 10^3)(2.872)}{10.5}$$

$$R = 18.04 \times 10^3 \Omega$$

$$\boxed{R = 18.04 \text{ k}\Omega}$$